CAROM BridgeLearn Shadow-Mode v2

Exact Experiment, Parameters, Complete Recorded Results, and Interpretation

ZeroBot research record for Benjamin Goertzel

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Abstract

This report documents the exact CAROM BridgeLearn v0.3.0 shadow-mode v2 calibration run. The run observed all 1,500 AdamW updates at matched per-update cadence while leaving the OneCycleLR schedule unchanged. It recorded parameter-variance transitions, split-batch noise estimates, random-direction Hessian-vector curvature probes, sharpness forecasts, and AR(2) dynamics for four parameter blocks. The key positive result is that one-step predicted and realized parameter variances agree almost perfectly (correlations 0.999996–0.999999), showing that the large discrepancy in the earlier cadence-mismatched run was mainly a horizon artifact. The active control gate nevertheless fails: predicted innovation is orders of magnitude below empirical transition innovation, sharpness intervals are extremely wide and near-zero-confidence, and the AdamW trajectory is stably second order rather than adequately AR(1). These failures are not all equivalent to ordinary scalar hyperparameter mistuning. Some parameters require calibration, but the innovation mapping and state model require structural changes before active BridgeLearn control is scientifically justified.

Contents

1 Question and status

The experiment asked:

> Does matched per-update observation, Hessian-vector-product sharpness, split-batch innovation measurement, and an AR(2) diagnostic make BridgeLearn sufficiently calibrated for an active CAROM control experiment?

Observed: The command exited with status 0. All 1,500 shadow non-mutation checks passed, all telemetry is finite strict JSON, and the run produced 1,500 per-update observations, 30 split-batch measurements, 1,498 available AR(2) fits, and 11 evaluation summaries.

Decision: The mechanical shadow integration passes. Activation does not: BridgeLearn must remain in shadow mode until innovation, sharpness, and momentum-state adequacy gates pass.

2 Exact provenance and environment

Item	Recorded value
Experiment ID	`20260722T200000Z-bridgelearn-shadow-mode-v2`
Experiment directory	`projects/carom/experiments/20260722T200000Z-bridgelearn-shadow-mode-v2/`
CAROM repository	`projects/carom/repos/carom`
Workspace branch	`agent/chat-room-identity-phase1`
Recorded workspace commit	`ffdc2934e8006a27ed9e0f7d0e7137f8456833fd`
Runner status	Runner and experiment were uncommitted in a shared dirty workspace; exact runner content is preserved by SHA-256.
Runner SHA-256	`9443bb973b7885131aaf6c3506403c408e3b427d5e10e17b5b04f105e38ac2c5`
BridgeLearn	Version 0.3.0, dedicated local virtual environment
Python	3.10.12
PyTorch	2.13.0+cpu
Platform	Linux x86_64, CPU only
Elapsed wall time	703.929 s in telemetry; 11:46.49 from `/usr/bin/time -v`
CPU utilization	461 percent aggregate
Maximum resident set	1,008,852 KiB
Exit status	0

This was not a paid or remote run. No RunPod resource was active.

3 Exact command

```
cd /home/openclaw/research-agent/projects/carom/repos/carom
/usr/bin/time -v \
  /home/openclaw/research-agent/scratch/bridgelearn-carom-venv/bin/python \
  run_carom_bridge_shadow_v2.py scheduled \
  --steps 1500 \
  --seed 0 \
  --device cpu \
  --output \
```

```
/home/openclaw/research-agent/projects/carom/experiments/\
20260722T200000Z-bridgelearn-shadow-mode-v2/telemetry.json
```

All unspecified command-line arguments took the runner defaults listed below.

4 Complete parameter specification

4.1 Task and model

Parameter	Exact value or behavior
Execution variant	`scheduled`; commands are applied sequentially in presentation order.
Workspace	Six slots over $\mathbf{Z}_8$.
Program length	Uniform integer from 2 through 5 for each example; padded with trailing no-ops to length 5.
Command vocabulary	Ten tokens: no-op, increment, double, negate, reverse, shift-right, swap-first-two, cumulative modular sum, increment synonym, and reverse synonym.
Input generation	Procedural; six independently uniform values in $\{0, \ldots, 7\}$ followed by exact execution of the sampled command list.
Model width	$d = 48$.
Operator count	$K = 8$.
Routing	Learned command embedding with K logits, softmax temperature $\tau = 1.0$.
Shared operator core	Eight parallel read-transform-gate-write operators, with attention, GELU transform, sigmoid gate, and residual workspace update.
Embeddings	Symbol embedding for nine values including blank; six position embeddings.
Readout	LayerNorm, linear $48 \to 96$, GELU, linear $96 \to 8$.
Parameter blocks	embeddings; routing; shared operator core; readout plus any otherwise unassigned parameters.
Training loss	Cross-entropy over all six output slots.
Training batch size	128.
Evaluation batch size	512, newly generated from the same procedural distribution.
Seed	0 for Python `random`, NumPy, and PyTorch.
HVP random seed	173 (training seed plus 173).

4.2 Optimizer and schedule

Parameter	Exact value or behavior
Optimizer	PyTorch AdamW.
Learning-rate argument / OneCycle maximum	2.0×10^{-3}.
OneCycleLR unspecified defaults	`pct_start=0.3`, `anneal_strategy='cos'`, `cycle_momentum=True`, `base_momentum=0.85`, `max_momentum=0.95`, `div_factor=25`, `final_div_factor=10000`, `three_phase=False`.
Initial LR	8.0×10^{-5}.
Peak LR	Approximately 2.0×10^{-3} near update 450.
Final recorded pre-update LR	8.0×10^{-9} at update 1499.
AdamW defaults not overridden	$\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, AMSGrad false, maximize false.
Weight decay	10^{-4}.
Gradient clipping	Global norm clipped to 1.0 before each optimizer update.

Parameter	Exact value or behavior
Updates	1,500.
Evaluation cadence	Every 150 updates and at final update.

4.3 BridgeLearn observer and planner parameters

Component	Exact parameters
Controller mode	`shadow`; proposed controls were recorded but not applied to optimizer or scheduler state.
Observation cadence	Before and after every optimizer update.
Empirical transition	Paired pre/post-update flattened parameter samples, separately for four blocks.
Split-batch noise	Batch divided into two halves every 50 updates; floor 10^{-12}; exponential filter $\alpha = 0.2$; latest estimate reused between measurements.
Sharpness probe	One independent random unit vector per block and update; absolute Rayleigh quotient $\lvert v^\top Hv \rvert$ computed with autograd HVP.
Sharpness filter	Exponential filter $\alpha = 0.1$.
Sharpness plant	Default BridgeLearn `SharpnessPlant`; consecutive filtered HVP values; progress signal fixed to one.
AR(2) batch observer	Rolling window 128 updates; 16 evenly spaced fixed parameter coordinates per block; minimum 24 samples.
AR(2) online cross-check	Forgetting-factor recursive least squares with forgetting 0.99.
AR(1) adequacy threshold	0.5.
Initial curvature	1.0 per block.
Initial noise scale	10^{-8} per block.
Initial AR(1) adequacy	0.5 per block.
Commanded contraction	$1 - \eta h(\eta)$.
Commanded innovation	$\max(\eta^2\,\text{noise/effective batch}, 10^{-15})$.
Reference filter	$\alpha = 0.03$; maximum contraction change 0.01; maximum log-innovation change 0.20.
Step actuator bounds	10^{-7} to 5×10^{-3}; maximum log-step change 0.15.
Batch actuator	Fixed at 128; quantum 128.
Planner terminal variance	Half the initial block variance, floored at 10^{-12}.
Planner horizon	1,500; maximum horizon multiplier 16; approximate progress permitted.
Probe policy	Confidence threshold 0.4; cold variance 0.01; maximum probe innovation 10^{-7}; maximum total probe 10^{-5}.
Momentum handling	First-order momentum permitted; AR(1) and structural diagnostics recorded but not required for shadow planning.

5 Evaluation trajectory

Update	LR	Train loss	Eval accuracy	Max |variance error|
0	8.000e-5	2.1001	0.1175	2.270e-5
150	5.619e-4	2.0440	0.1810	1.916e-4
300	1.524e-3	1.9784	0.2012	4.148e-4
450	2.000e-3	1.8199	0.2497	1.055e-3
600	1.900e-3	1.7245	0.3171	8.795e-4
750	1.621e-3	1.5875	0.3275	4.829e-4
900	1.220e-3	1.5415	0.3730	3.937e-4
1050	7.746e-4	1.4008	0.4141	2.586e-4
1200	3.742e-4	1.4253	0.4346	1.613e-4
1350	9.774e-5	1.3387	0.4424	3.259e-5
1499	8.000e-9	1.4378	0.4137	1.475e-8

Across all 1,500 stochastic training batches, mean loss was 1.66668, minimum loss was 1.19589, mean training accuracy was 0.31809, and maximum single-batch training accuracy was 0.52474. These summaries are descriptive only: the run was designed to calibrate observers, not to estimate final model efficacy.

6 Transition and variance calibration

Block	Bias	MAE	RMSE	Max abs. error	Correlation
Embeddings	−4.63e-5	6.58e-5	1.03e-4	5.33e-4	0.9999961
Routing	+3.74e-4	3.74e-4	4.84e-4	1.24e-3	0.9999982
Operator core	+2.62e-6	2.62e-6	3.62e-6	1.04e-5	0.9999988
Readout	+1.19e-5	1.21e-5	1.60e-5	5.08e-5	0.9999988

The residual 5th, 50th, and 95th percentiles were:

Block	q05	q50	q95
Embeddings	−2.27e-4	−1.71e-5	+6.05e-5
Routing	+2.63e-6	+3.03e-4	+9.30e-4
Operator core	+1.30e-8	+1.96e-6	+7.68e-6
Readout	−6.01e-7	+1.08e-5	+3.13e-5

Contraction MAEs for embeddings, routing, operator core, and readout were, respectively, 3.19×10^{-5}, 1.76×10^{-4}, 7.42×10^{-4}, and 2.68×10^{-4}. The maximum variance error occurred in routing at update 444, near the OneCycleLR peak.

Inferred: The earlier v1 maximum variance error of 0.111 and contraction gap of 0.338 were primarily caused by comparing one-step predictions with realized statistics separated by 149–150 optimizer updates. At matched horizon the variance model is exceptionally accurate. The small positive routing bias is real in this trace but is not the main activation blocker.

7 Innovation mismatch

Block	Median predicted	Median empirical	Median split-batch unit noise
Embeddings	1.09e-12	9.25e-8	1.96e-4
Routing	2.07e-11	3.07e-7	6.17e-3
Operator core	5.11e-14	1.06e-7	1.15e-5
Readout	1.09e-11	6.62e-8	1.08e-3

Mean absolute predicted-versus-empirical innovation gaps were 2.25e-7, 7.52e-7, 2.16e-7, and 1.39e-7 in block order.

Observed: The controller's current innovation estimate is several orders of magnitude smaller than the empirical parameter-transition innovation.

Hypothesis: Possible contributors are the current η^2/B conversion, AdamW preconditioning, momentum, weight decay, the split-batch estimator's scale, and the fact that empirical innovation is inferred across parameter coordinates rather than from replicated stochastic trajectories beginning at the same parameter state.

Inferred: This is not yet identifiable as a single scalar tuning problem. A replicated same-state transition experiment is needed to determine whether a constant scale correction suffices or whether the mapping must model optimizer state and blockwise covariance.

8 Sharpness estimates and forecast calibration

Block	Filtered HVP range	Center MAE	Median relative width	Final confidence
Embeddings	2.04e-5–6.99e-3	9.84e-5	73x	5.47e-7
Routing	1.60e-6–9.77e-2	1.59e-3	3.70x	5.47e-7
Operator core	4.94e-6–8.28e-4	1.81e-5	537x	5.47e-7
Readout	1.88e-3–1.49e-2	2.27e-4	16.0x	5.47e-7

Nominal empirical interval coverage was 99.8–100 percent. This high coverage is not evidence of a useful forecast because the intervals are extremely wide and confidence is essentially zero. The v1 gradient-secant values were many orders larger than the v2 HVP values; the secant proxy should not be treated as Hessian sharpness.

Hypothesis: This component may admit parameter improvement—for example plant warm-start, regularization, or better filter constants—but the single random Rayleigh direction does not estimate the largest curvature eigenvalue. Multiple HVP probes or a short power iteration is a structural estimator upgrade, not merely parameter tuning.

9 AR(2) dynamics and the AR(1) failure

AR(1) adequacy was zero at threshold 0.5 for every block on every available fit. Mean adequacy scores were 1.18×10^{-8}, 4.00×10^{-10}, 8.67×10^{-9}, and 6.41×10^{-9}.

Block	Final ϕ_1	Final ϕ_2	Spectral radius	Stable-fit fraction
Embeddings	1.631	-0.641	0.971	97.8%
Routing	1.818	-0.823	0.971	90.5%
Operator core	1.567	-0.579	0.970	100.0%
Readout	1.518	-0.528	0.979	99.2%

The fitted dynamics are mostly stable under the Jury criterion; this is not a finding of explosive optimization. The negative second lag and spectral radii just below one are consistent with persistent optimizer momentum.

Inferred: A scalar AR(1) model is structurally mismatched to this AdamW trajectory. Tuning its coefficient cannot in general represent a stable AR(2)-like process. A companion-state or explicit momentum adapter is the direct repair.

10 What "parameter tuning issue" means here

The data divide the problem into three categories:

1. **Already calibrated after measurement repair.** One-step parameter-variance prediction is highly accurate. No major retuning is indicated for this component, although routing has a small positive bias.
2. **Potential parameter and estimator calibration.** Sharpness EMA, plant warm-start, confidence regularization, number of HVP probes, and probe policy thresholds should be swept in shadow mode. Some are scalar parameters; the move from one random direction to an eigenvalue-oriented estimate changes the estimator itself.
3. **Structural mismatch before tuning.** The innovation conversion has not been calibrated against replicated same-state stochastic updates, and the AR(1) state model omits AdamW momentum. Sweeping controller gains on top of these mismatches risks obtaining apparent calibration for the wrong reason.

Decision: The next experiment should remain shadow-only and should first identify the innovation map and momentum state. Only after those are fixed should ordinary controller hyperparameters be tuned for active operation.

11 Recommended bounded next gate

1. Freeze selected parameter states at early, peak-LR, descending-LR, and terminal schedule phases.
2. From each state, execute replicated stochastic updates with identical parameters and optimizer state but independent batches. Estimate blockwise innovation covariance directly.
3. Fit and test candidate mappings: scalar rescale; η^2/B with AdamW-preconditioned gradient covariance; and a companion-state model that includes first-moment momentum.
4. Compare AR(1), AR(2), and explicit AdamW-state predictions on held-out update intervals using the same per-update cadence.
5. Replace the one-direction curvature probe with multiple probes or short power iteration; tune EMA and plant regularization only after estimator repeatability is quantified.
6. Require held-out calibration error, useful interval width, nontrivial confidence, stability margins, and strict LR non-mutation before considering active control.

This sequence discriminates true parameter mistuning from model misspecification without another expensive training run.

12 Limitations

- One seed and one CAROM scheduled variant were tested.
- The training and evaluation data are procedurally sampled from the same distribution; this is an observer-calibration run, not a generalization study.
- The runner was uncommitted in a shared dirty workspace. Its exact hash is preserved, but commit-only reproduction is unavailable.
- Sharpness uses one random direction per block per update and absolute Rayleigh quotient; it is neither a trace estimate averaged over probes nor a largest-eigenvalue estimate.
- Split-batch and empirical innovations are not measurements of precisely the same stochastic object.
- AR(2) uses 16 fixed coordinates per block rather than the full parameter state.
- Evaluation points use fresh stochastic batches, so small endpoint accuracy changes should not be overinterpreted.
- BridgeLearn proposed controls were not applied. This run provides no evidence that active BridgeLearn control improves CAROM accuracy or stability.

13 Telemetry field dictionary

Every per-update record contains:

`step`, `loss`, `training_accuracy`
Update index and current stochastic-batch outcomes.

`lr`, `lr_after_scheduler`
Four block learning rates before and after the OneCycleLR scheduler step.

`observed_variance`, `predicted_variance`, `calibration_error`
Per-block realized parameter population variance, BridgeLearn forecast, and realized minus predicted value.

`predicted_transition`, `realized_transition`
Per-block contraction and innovation objects.

`sharpness_raw`, `sharpness_estimate`
Raw and EMA-filtered absolute random-direction HVP Rayleigh values.

`sharpness_funnel`
Lower, center, upper, confidence, realized error, and coverage for the forecast made on the prior update.

`split_batch_innovation`
Per-block observation every 50 updates; null on intervening updates.

`ar2_adequacy`
Availability, ϕ_1, ϕ_2, innovation, spectral radius, AR(1) residual autocorrelation, adequacy score, regressor condition, Jury margins, stability verdict, and online RLS cross-check.

`bridge_proposed_controls`
Controls proposed by BridgeLearn but not applied.

`shadow_lr_invariant`
Exact per-update Boolean that the shadow apply did not mutate LR.

`grad_norm`
Global gradient norm before clipping to 1.0.

The top-level telemetry also records schema and BridgeLearn versions, mode, variant, seed, device, steps, batch sizes and cadences, block labels, elapsed seconds, evaluation summaries, calibration report, control effort, final evaluation accuracy, and serialized sharpness-plant state. The complete 1,500-row, approximately 8.2 MB JSON is the authoritative raw-data appendix; reprinting it verbatim inside the PDF would reduce usability without adding information.

14 Artifact verification

Artifact	SHA-256
`telemetry.json`	`0b9b18d6b91452cdfc15928dea79ad7d39da3e2d3f23cdb7cc7460a32c25dc2c`
`stdout.log`	`609c94ad70151409e54c17fc36debde6042d97b4ee19b04c0ccb14daab49af7c`
`stderr.log`	`93cb210d703a0853e0ed14f7c4575c088a97dcff519184e35585631617bba6ec`
`command.sh`	`618577204571c126f62e0a79d52ee8e03df9c0bfacc9b8a4f5b5cdc04fb28ead`
`run_carom_bridge_shadow_v2.py`	`9443bb973b7885131aaf6c3506403c408e3b427d5e10e17b5b04f105e38ac2c5`

15 Bottom line

Observed: Matching the prediction and observation horizon removed the dramatic v1 variance anomaly. BridgeLearn follows one-step block variance extremely well, even near peak OneCycleLR.

Observed: Innovation calibration, sharpness informativeness, and AR(1) adequacy still fail.

Inferred: The remaining problem is partly tunable but not purely a conventional hyperparameter issue. The evidence specifically favors (i) direct same-state innovation identification, (ii) an explicit momentum/AR(2) state model, and (iii) a repeatable multi-probe curvature estimator, followed by bounded shadow-mode tuning.

Decision: Active control should not begin from the present scalar AR(1) configuration.